

LEARNER WORKBOOK

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	ACHIEVED	NOT ACHIEVED
KNOWLEDGE	☒	
SKILLS	☐	

Well Done

S.L. Mawasha.

Courtney Rosenberg

Signature of learner

Signature of Assessor

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Learner workbook

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This material has been compiled based on the **UNIT STANDARD 9010: Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations.**

The program has been accredited by SASSETA in the Republic of South Africa.

Material compiled by: Training Dynamics Africa

HOW THIS LEARNER WORKBOOK IS COMPILED

This unit standard is designed to provide credits towards the mathematical literacy requirements of the NQF at level 2. The essential purposes of the mathematical literacy requirements are that, as the learner progresses with confidence through the levels, the learner will grow in:

- ♦ An insightful use of mathematics in the management of the needs of everyday living to become a self-managing person.
- ♦ An understanding of mathematical applications that provides insight into the learner's present and future occupational experiences and so develop into a contributing worker.
- ♦ The ability to voice a critical sensitivity to the role of mathematics in a democratic society and so become a participating citizen.

People credited with this unit standard are able to:

- ♦ Convert numbers between the decimal number system and binary number system.
- ♦ Work with numbers in different ways to express size/magnitude.
- ♦ Demonstrate the effect of error in calculations.

You have to study the content and guidelines given in your learner reference guide to complete this workbook.

Remember that this completed workbook must be handed in as it will form part of your formative assessment and will play a role in awarding you the required 05 credits that are to be achieved upon completion of this module.

Write neatly and legibly and ensure that you do not simply copy the information contained in the learner reference guide into your workbook. Give your own opinion and ideas where applicable and write the answers in your own words to indicate your level of understanding and learning.

Good luck with the completion of this workbook.

LEARNING UNIT 1

Number systems and conversions

LEARNING OUTCOMES

A learning credited with this unit will be able to:

- ♦ Represent numbers in different ways.
- ♦ Round different types of numbers.
- ♦ Work with rational and irrational numbers and proper and improper fractions.
- ♦ Perform arithmetic with fractions.
- ♦ Calculate error ranges that result from arithmetic with rounded numbers.
- ♦ Estimate the correct answer to an arithmetic problem.
- ♦ Convert numbers between the decimal and binary systems.
- ♦ Perform addition and subtraction with binary numbers.
- ♦ Understand and work with the SI system of units.
- ♦ Understand the relationship between the SI system and other number systems.
- ♦ Estimate sizes of objects in the SI and UK systems.

LEARNER ACTIVITY 1

Using the columns given, convert the following numbers from binary to decimal:

♦ 111_2

⁶ 2	⁵ 2	⁴ 2	³ 2	² 2	¹ 2	⁰ 2
64's	32's	16's	8's	4's	2's	1's
				1	1	1

$=4+2+1=7$



♦ 11001_2

⁶ 2	⁵ 2	⁴ 2	³ 2	² 2	¹ 2	⁰ 2
64's	32's	16's	8's	4's	2's	1's
		1	1	0	0	1

$=16+8+0+0+1=25$



♦ 101_2

⁶ 2	⁵ 2	⁴ 2	³ 2	² 2	¹ 2	⁰ 2
64's	32's	16's	8's	4's	2's	1's
				1	0	1

$=4+0+1=5$



Requirements achieved	YES			NO	
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Comments in support of decision

Achieved

UNIT STANDARD 9010: Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations.

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Requirements achieved	YES	☒	☐	NO
Comments in support of decision				

Achieved

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LEARNER ACTIVITY 3

Add the following binary numbers: (Convert these binary numbers to decimal numbers at the end and check to see if the addition is correct)

♦ $1011_2 + 101_2$

$$1011_2$$

$$+ 101_2$$

$$10000_2$$


☐

b.) Convert each of the binary numbers to decimal.

64	32	16	8	4	2	1
			1	0	1	1

$$= 8 + 0 + 2 + 1 = 11$$

$$= 1011_2$$

64	32	16	8	4	2	1
				1	0	1

$$= 4 + 0 + 1 = 5$$

$$= 101_2$$

$$= 16 + 0 + 0 + 0 + 0 = 16$$

64	32	16	8	4	2	1
		1	0	0	0	0

0

$$10000_2 = 16$$

$$= 1000$$


☐

Checking to see if the addition is correct

$$1011_2 + 101_2 = 10000_2$$

$$11 + 5 = 16$$

$$16 = 16$$

The two sides are equal, therefore the addition is correct.


☐

UNIT STANDARD 9010: Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations.

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Requirements achieved	YES	☒	☐	NO
Comments in support of decision				

Achieved

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LEARNER ACTIVITY 4

Add the following binary numbers: (Convert these binary numbers to decimal numbers at the end and check to see if the addition is correct)

♦ $1001_2 + 1100_2$

$$\begin{array}{r} 1001_2 \\ + 1100_2 \\ \hline = 10101_2 \end{array}$$


☐

i). Convert each of the binary numbers to decimal numbers.

$1001_2 =$

64	32	16	8	4	2	1
			1	0	0	1 ₂

$1001_2 = 8 + 0 + 0 + 1$

$1001_2 = 9$


☐

ii). $1100_2 =$

64	32	16	8	4	2	1
			1	1	0	0 ₂

$1100_2 = 8 + 4 + 0 + 0$

$1100_2 = 12$

iii). $10101_2 =$

64	32	16	8	4	2	1
		1	0	1	0	1 ₂

$10101_2 = 16 + 0 + 4 + 0 + 1$

$10101_2 = 21$

Checking to see if the additions are correct.


☐

$1001_2 + 1100_2 = 10101_2$

$9 + 12 = 21$

$21 = 21$

The two sides are equal, therefore the additions are correct.

UNIT STANDARD 9010: Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations.

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Requirements achieved	YES	☒	☐	NO
Comments in support of decision				

Achieved

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Ronny



LEARNER ACTIVITY 5

Add the following binary numbers: (Convert these binary numbers to decimal numbers at the end and check to see if the addition is correct)

♦ $1100_2 + 1101_2$

$$\begin{array}{r} 1100_2 \\ + 1101_2 \\ \hline 11001_2 \end{array}$$



Converting each binary number to decimal number:

i). $1100_2 =$

64	32	16	8	4	2	1
			1	1	0	0 ₂

$$1100_2 = 8 + 4 + 0 + 0$$

$$\underline{1100_2 = 12}$$



ii). $1101_2 =$

64	32	16	8	4	2	1
			1	1	0	1 ₂

$$1101_2 = 8 + 4 + 0 + 1$$

$$\underline{1101_2 = 13}$$

iii). Checking to see if the addition is correct:

$$1100_2 + 1101_2 = 11001_2$$

$$1100_2 + 1101_2 = 12 + 13$$

$$12 + 13 = 25$$

$$\underline{25 = 25}$$



The two sides are equal, therefore the addition is correct.

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[illegible]

Requirements achieved	YES		☐	NO
Comments in support of decision				

Achieved

LEARNER ACTIVITY 6

Subtract the following binary numbers: (Convert these binary numbers to decimal numbers at the end and check to see if the addition is correct)

♦ $1011_2 - 101_2$

$$1011_2 - 101_2 = \begin{array}{r} 1011_2 \\ - 101_2 \\ \hline = 110_2 \end{array}$$



Converting each binary number including the total to decimal numbers:

i). $1011_2 =$

64	32	16	8	4	2	1
			1	0	1	1 ₂

$1011_2 = 8 + 0 + 2 + 1$

$1011_2 = 11$



ii). $101_2 =$

64	32	16	8	4	2	1
				1	0	1 ₂

$101_2 = 4 + 0 + 1$

$101_2 = 5$

iii). $110_2 =$

64	32	16	8	4	2	1
				1	1	0 ₂

$110_2 = 4 + 2 + 0$

$110_2 = 6$



Check to see if the subtraction is correct:

$1011_2 - 101_2 = 110_2$

$11 - 5 = 6$

$6 = 6$

The two are equal, therefore the subtraction is correct.

LEARNER ACTIVITY 7

Subtract the following binary numbers: (Convert these binary numbers to decimal numbers at the end and check to see if the addition is correct)

♦ $1100_2 - 1001_2$

$$\begin{array}{r} 1100_2 - 1001_2 = 1100_2 \\ - 1100_2 \\ \hline = 110_2 \end{array}$$



Converting each binary number including the difference to decimal numbers:

i). $1100_2 =$

64	32	16	8	4	2	1
			1	1	0	0 ₂

$$1100_2 = 8 + 4 + 0 + 0$$

$$\underline{1100_2 = 12}$$

ii). $1001_2 =$

64	32	16	8	4	2	1
			1	0	0	1 ₂

$$1001_2 = 8 + 0 + 0 + 1$$

$$\underline{1001_2 = 9}$$

iii). $11_2 =$

64	32	16	8	4	2	1
					1	1_2

$$11_2 = 2 + 1$$

$$11_2 = 3$$

$$1001_2 - 1001_2 = 11_2$$

$$12 - 9 = 3$$

$$\underline{33}$$

The two are equal, therefore the subtraction is correct.

Requirements achieved	YES	☒	☐	NO
Comments in support of decision				

Achieved

LEARNER ACTIVITY 8

Subtract the following binary numbers: (Convert these binary numbers to decimal numbers at the end and check to see if the addition is correct)

♦ $10000_2 - 1010_2$

$$\begin{array}{r} 10000_2 \\ - 1010_2 \\ \hline 110_2 \end{array}$$



Converting each binary number including the difference to decimal numbers:

i). 10000_2

64	32	16	8	4	2	1
		1	0	0	0	0

$10000_2 = 16 + 0 + 0 + 0 + 0$

$10000_2 = 16$

ii). 1010_2

64	32	16	8	4	2	1
			1	0	1	0

$1010_2 = 8 + 0 + 2 + 0$

$1010_2 = 10$

iii). 110_2

64	32	16	8	4	2	1
				1	1	0

$110_2 = 4 + 2$

$110_2 = 6$

Checking to see if the subtraction is correct:

$$10000_2 - 1010_2 = 110_2$$

$$16 - 10 = 6$$

$$\underline{6 = 6}$$

The two sides are equal, therefore the subtraction is correct.

LEARNER ACTIVITY 10

Convert the following numbers from scientific to decimal notation:

- a) $1,23 \times 10^6$
b) $5,16 \times 10^{-3}$



a). 10^6 means 1000 000

Multiplying $1,23 \times 1000\ 000$ means shifting 6 steps to the right from the comma.

$$1,23 \times 10^6 = 123\ 0000$$

b). $5,16 \times 10^{-3}$ (10^{-3} means $\frac{1}{1000}$)

$$5,16 \times 10^{-3} \text{ becomes } \frac{5,16 \times 1}{1000}$$

Dividing 5.16 by 1000 means shifting 3 places to the left from the comma.

$$5,16 \times 10^{-3} = 0,00516$$

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Requirements achieved	YES		☐	NO
Comments in support of decision				

Achieved

LEARNER ACTIVITY 11

Convert the following measurements as instructed, writing your answers in scientific notation where necessary.

- a) 12 km to m
- b) 20 cm to m
- c) 4,65 km to m
- d) 5 345 mm to m
- e) 125 litres to kilolitres
- f) 0,0073 grams to pictograms

a) $12 \text{ km to m} = 12 \text{ km} \times 1000 = 1,2 \times 10^4 \text{ m.}$

b) $0,2 \text{ m} = 2,0 \times 10^1 \text{ m.}$

c) $4,560 \times 10^3 = 4560 \text{ m.}$

d) $5,345 \text{ m.}$

e) $0,125 \text{ Kl.} = 1,25 \times 10^{-1} \text{ Kl.}$

f) $0,0073 \times 10^{-12} = 7,3 \times 10^{-15} \text{ pg.}$



LEARNER ACTIVITY 12

- ♦ The boiling point of water is 100°C. You have a tank that has a temperature control in degrees Fahrenheit. What must you set the temperature to in order to boil the water in the tank?
- ♦ Danny is an operator at a bottling plant. Each bottle has a volume of 500 ml. He is given an export order to the USA for 400 gallons of vinegar.
 - i) How many bottles must he fill?
 - ii) Danny's colleague checks his calculations, but he rounds off the conversion rate to 3,8 litres. How many extra bottles would he have filled?
 - iii) The sales manager who took the order has rounded off the conversion rate to the nearest litre. How many bottles has he quoted for?

❖ Converting degree Celsius to degree Fahrenheit:

$$\begin{aligned}\text{Formula: } & (1,8 \times C) + 32 \\ & C = \text{degree Celsius} \\ & (1,8 \times 100) + 32 \\ & 180 + 32 \\ & = \underline{212^\circ\text{F}}\end{aligned}$$



❖ Converting gallon to litre:

$$1 \text{ gallon} = 4,54609 \text{ litre}$$

Converting 400 gallons to litres, multiply by 4,54609 litre

$$\begin{aligned}400 \times 4,54609 \\ = \underline{1818,436 \text{ litres}}\end{aligned}$$



i. How many bottles must he fill of 500ml?

Step 1: Convert 1818,436 litres to ml by multiplying by 1000

$$\Rightarrow 1818,436 \times 1000 = 1818436\text{ml.}$$



Divide by 500ml to see how many bottles must be filled:

$$\begin{array}{rcl}\underline{1818436} & & 3636,872 \text{ bottles} \\ 500 & = & \underline{3637 \text{ bottles}}\end{array}$$

ii. 3.8 Litre X 400 gallons = 1520 litres

Converting to ml. multiply X 1000

=> 1520 000ml. divide by 500ml. to see how many bottles.

=> 1520 000

500

= 3040 bottles.


☐

iii. Rounding off the conversion to the nearest litre:

4,54609 litre to the nearest litre = 5 litres.

To see how many bottles he quoted for:-

400 gallons X 5 litres

= 2 000 litres


☐

Converting to ml. for easy division by 500 ml. multiply by 500 ml. multiply by 1000

2 000 X 1000 = 2000 000 ml.

Divide by 500 ml. to see how many bottles.

2000 000

500

= 4 000 bottles

Requirements achieved	YES		☐	NO
Comments in support of decision				

Achieved

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1	Represent numbers in different ways.			☒
2	Round different types of numbers.			☒
3	Work with rational and irrational numbers and proper and improper fractions.			☒
4	Perform arithmetic with fractions.			☒
5	Calculate error ranges that result from arithmetic with rounded numbers.			☒
6	Estimate the correct answer to an arithmetic problem.			☒
7	Convert numbers between the decimal and binary systems.			☒
8	Perform addition and subtraction with binary numbers.			☒
9	Understand and work with the SI system of units.			☒
10	Understand the relationship between the SI system and other number systems.			☒
11	Estimate sizes of objects in the SI and UK systems.			☒



Evidence Collector Signature

29.05.2021

Date

S.L. Mawasha

Security Officer Signature

26.05.2021

Date